SIMULATION AND ANALYSIS OF PENTACENE-BASED ORGANIC FIELD EFFECT TRANSISTOR

Research Innovation Centre (RIC), Defence Research and Development Organization (DRDO)
IIT Madras Research Park

DR. E VARADARAJAN, *CLUSTER HEAD, RIC, DRDO, IITMRP*

SAIKANTH DACHA, *INDIAN INSTITUTE OF TECHNOLOGY MADRAS*

T SANDEEP KRISHNA, *INDIAN INSTITUTE OF TECHNOLOGY MADRAS*

V VARUN KUMAR, *INDIAN INSTITUTE OF TECHNOLOGY MADRAS*

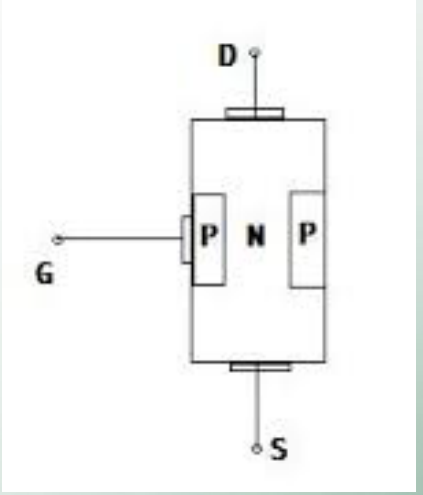
Summer Internship Final Presentation: 26 Aug'15

OUTLINE

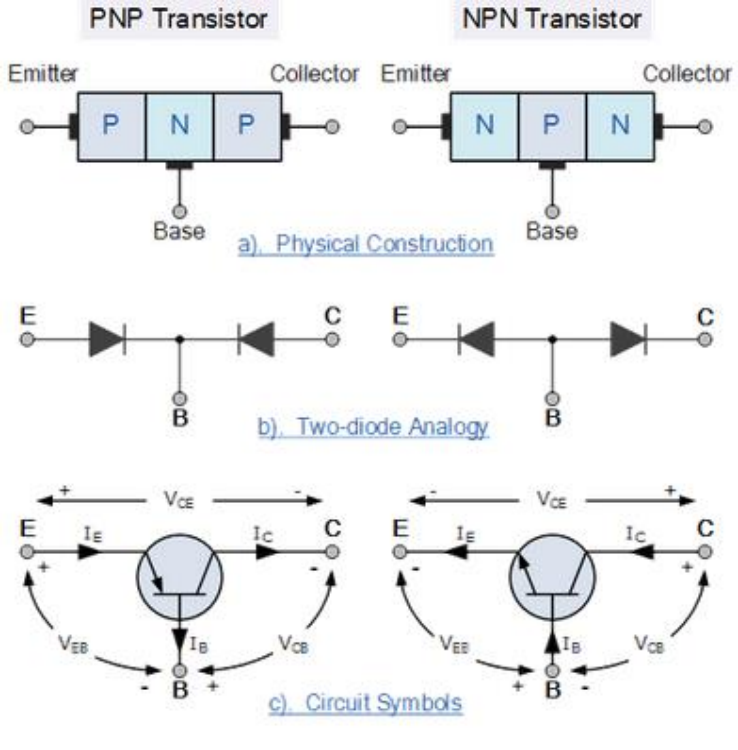
- Introduction to Transistors: BJT, MOSFET
- Motivation for our work
- Aim
- Introduction to Organic Semiconductors
- Pentacene
- More on Pentacene: Properties and Study on Mobility
- The Organic Field Effect Transistor (OFET)
- Device structure
- Simulation tools
- Simulation results
- Concluding remarks, Future work

INTRODUCTION

- Transistors:
  - Three terminal devices, generally made of semiconductor materials
  - Can act either as an insulator or a conductor by the application of a small signal voltage
  - Switching and amplification.



- Bipolar Junction Transistors (BJT) and Field effect Transistors (FET) are two popular transistors used widely.
- However, in the recent past, Organic Field effect Transistors (OFET) have gained much attention because of their advantages.



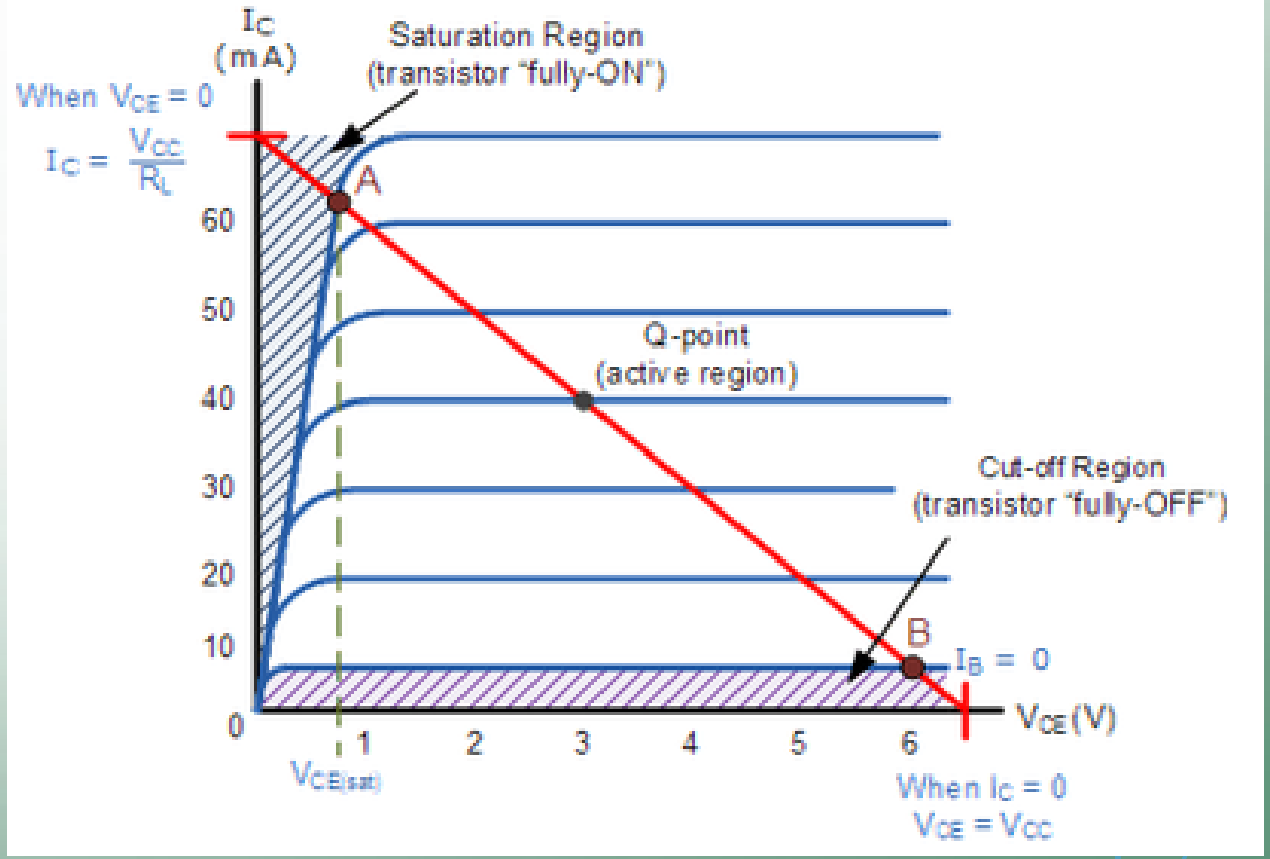
THE BIPOLAR JUNCTION TRANSISTOR

- A type of transistor that relies on the contact of two types of semiconductor for its operation.
- Invented in December 1947 by John Bardeen and Walter Brattain at the Bell Telephone Laboratories.
- Have the ability to operate within three different regions:

1) Active Region: The transistor operates as an amplifier,

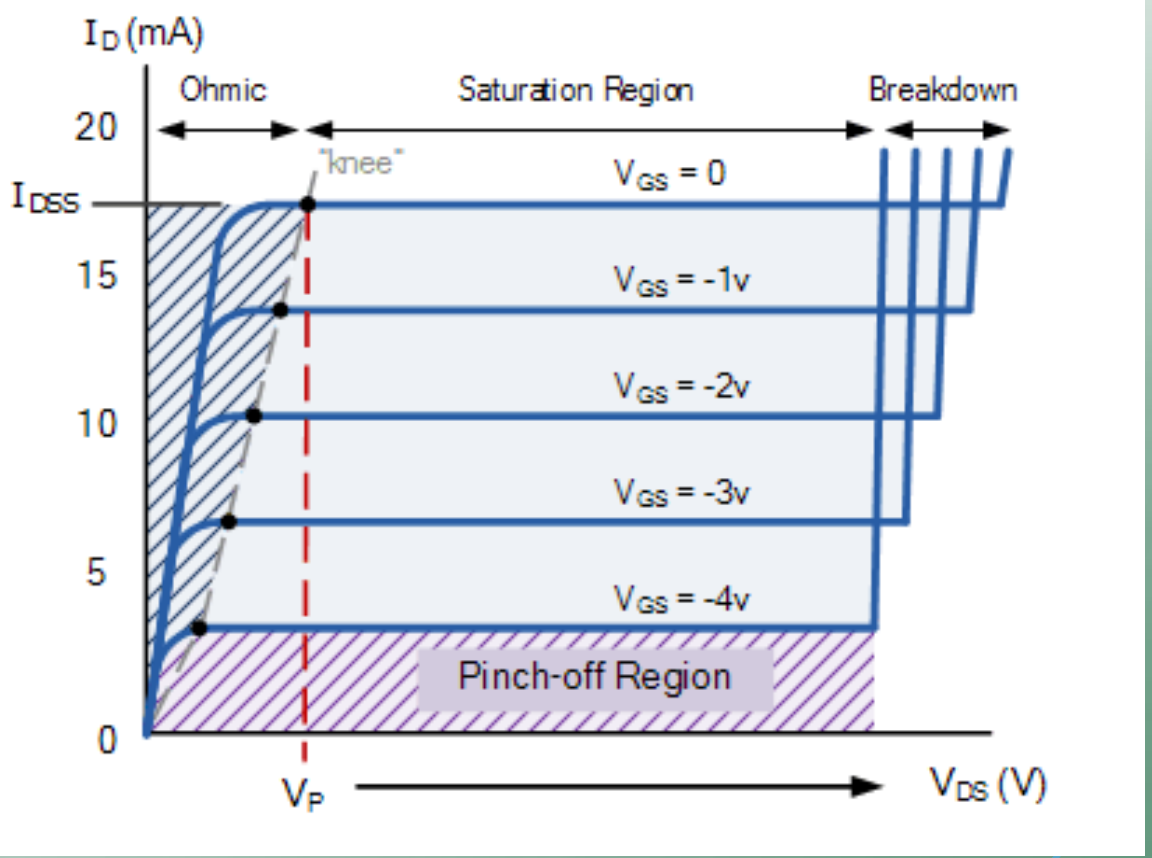
2) Saturation Region: The transistor is fully ON operating as a switch, and

3) Cut-off Region: The transistor is fully OFF operating as a switch.



THE FIELD EFFECT TRANSISTOR

- Uses an electric field to control the shape and hence the electrical conductivity of a channel
- Also known as unipolar transistors as they involve single-carrier-type operation.
- Used in CMOS process technology which forms the basis for modern digital integrated circuits.
- Operates in 4 different regions: *Cut-off Region, Ohmic/Linear Region, Saturation Region, Breakdown Region.*
- Triode region:
  $I_D = \mu_n C_{ox}(W/L)((V_{GS} - V_T)^2 - V^2_{DS}/2)$
  when $V_{GS} > V_T$ and $V_{DS} < (V_{GS} - V_T)$
- *Saturation region:*
  $I_D = \mu_n C_{ox}(W/2L)(V_{GS} - V_T)^2(1 + \lambda (V_{DS} - V_{DS,sat}))$
  when $V_{GS} > V_T$ *and* $V_{DS} > (V_{GS} - V_T)$



MOTIVATION

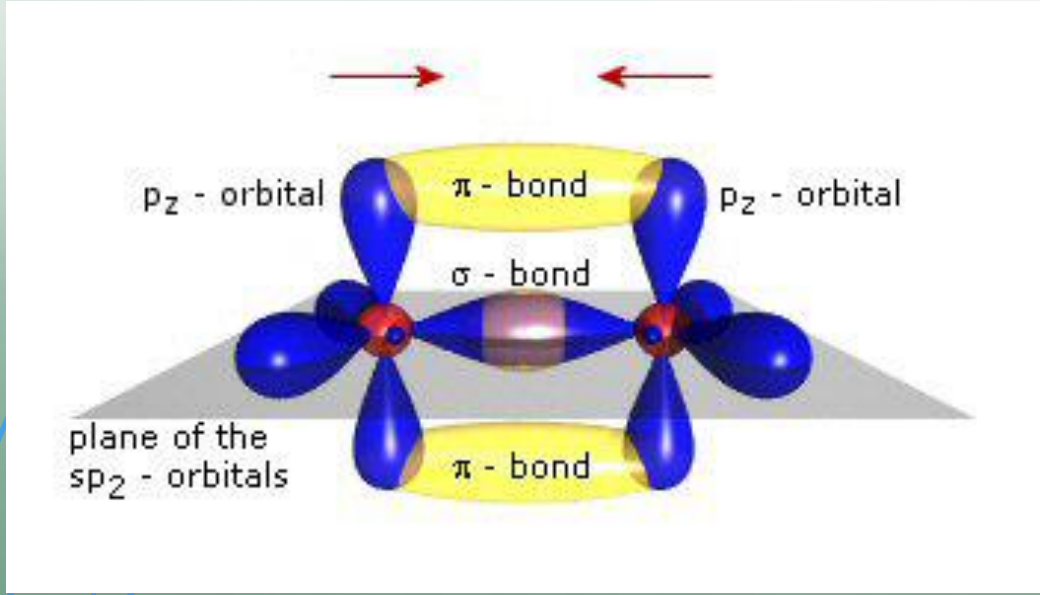
- Organic semiconductors (OSCs): low-cost, flexible, lightweight, and environmentally-friendly semiconductors.
- Hole mobility was found to be comparable to that of amorphous Silicon (a-Si).
- Applications in organic light-emitting diodes (OLEDs), photo-voltaic cells, and organic field-effect transistors (OFETs).

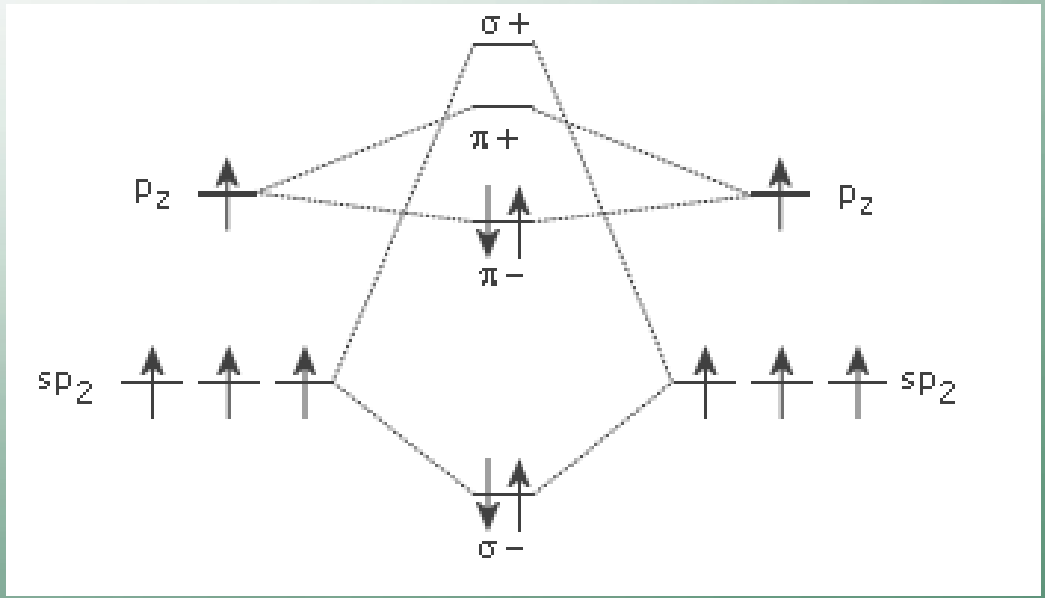
AIM

- Study transistor working: MOSFET, OFET
- Come up with device structures based on experiments from literature
- Simulate Pentacene-based OTFT's using Silvaco's TCAD tools to study the device characteristics

ORGANIC SEMICONDUCTORS

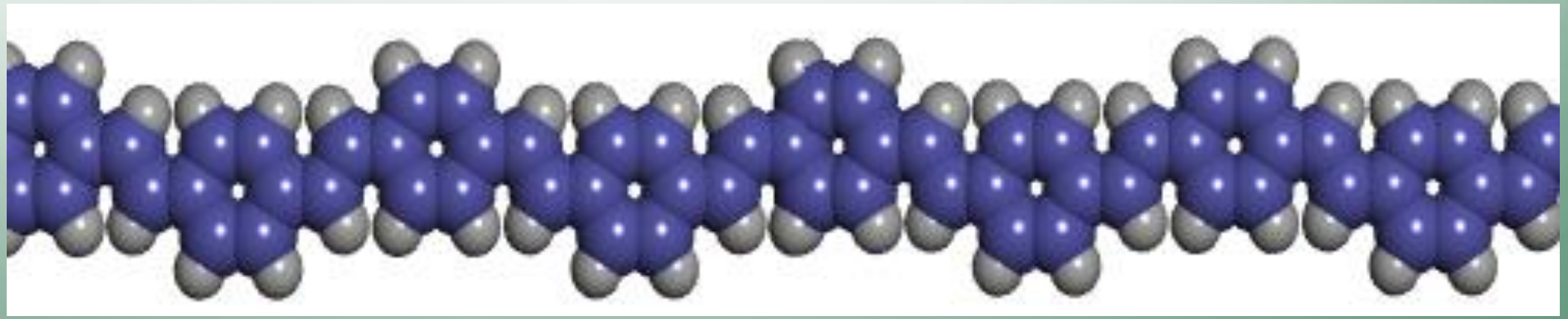
- An organic material with semiconductor properties.
- The semiconducting properties are based on the properties of the carbon atoms.
- *The bonding of sp2-hybridised carbon*: Small difference between the HOMO and LUMO, giving rise to semiconducting properties.





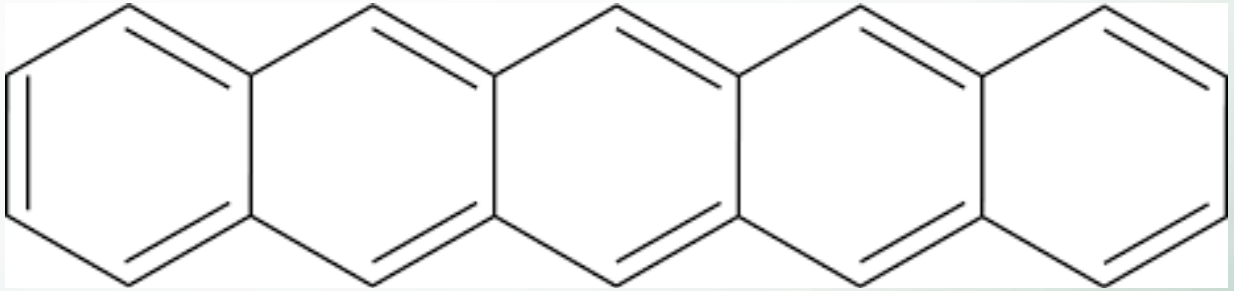
ORGANIC SEMICONDUCTORS

- *Small-molecule Organic Semiconductors:* If Carbon atoms form large molecules the pi-bonds become delocalized showing remarkable transport properties.
- *Polymer Organic Semiconductors:* Due to delocalization of pi-bonds, form a one-dimensional electronic system.



PENTACENE

- Pentacene is a flat molecule made of five linearly fused benzene rings, as shown in the figure below.



- Highly conjugated compound; works as an organic semiconductor
- Crystals and thin films behave as p-type semiconductors which can be used in the manufacturing of OFET/OTFT

PENTACENE

- Some properties of Pentacene in comparison with silicon

Property	Pentacene	Silicon
Bandgap	2.2 eV	1.1eV
Density of conduction band	$2 * 10^{21}$ cm^{-3}	$2.8 * 10^{19}$ cm^{-3}
Density of valence band	$2 * 10^{21}$ cm^{-3}	$1.0 * 10^{19}$ cm^{-3}
Permittivity	4.0	11.68
Affinity	2.8eV	4.05eV

MOBILITY

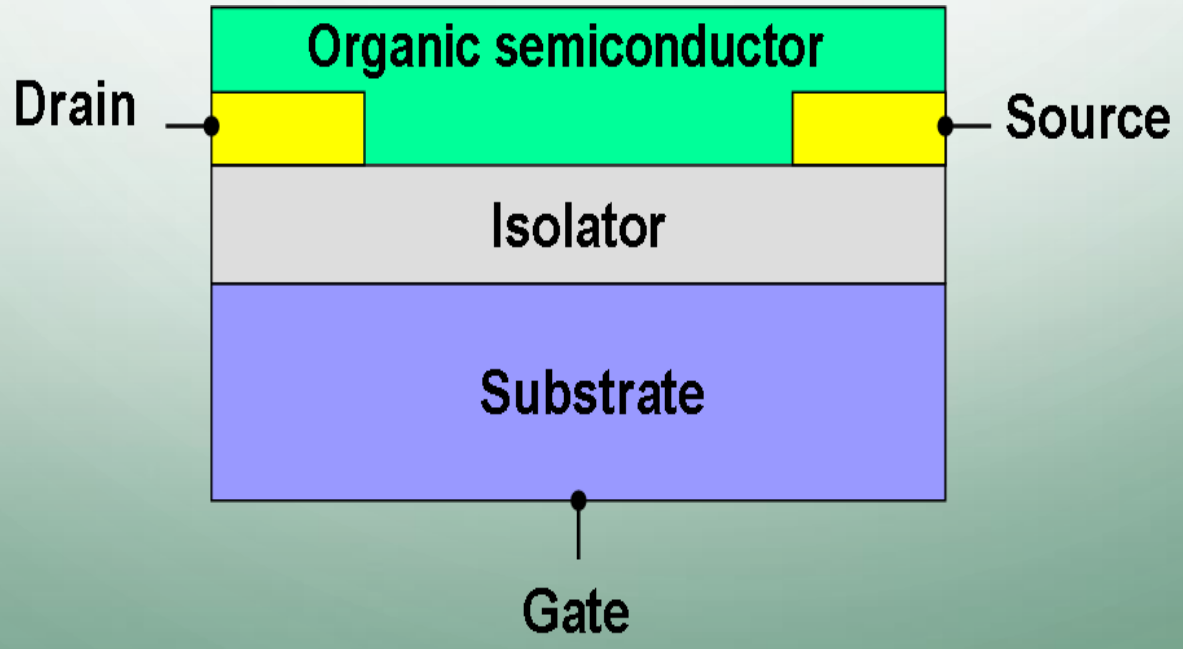
- Pentacene FETs:
  - Usually operate as p-channel (hole transporting) transistors
  - Have lesser mobility values compared to the current generation FETs
  - Initial mobility value of Pentacene FETs is 0.002 $cm^2/(V.s)$ [Horowitz et. Al., 1991]
- By using surface treatment of gate dielectrics, high mobilities of more than 1 $cm^2/(V.s)$ have been demonstrated

MOBILITY

- Mobilities of 3.0 , 3.3 and 5.5 $cm^2/(V.s)$ were reported in 2002, 2003 and 2006 respectively
- Methods for improving performance: Purification, suitable Grain size and Valleys, Surface Treatment, suitable gate dielectrics, Drain and Source electrodes

ORGANIC FIELD EFFECT TRANSISTOR (OFET)

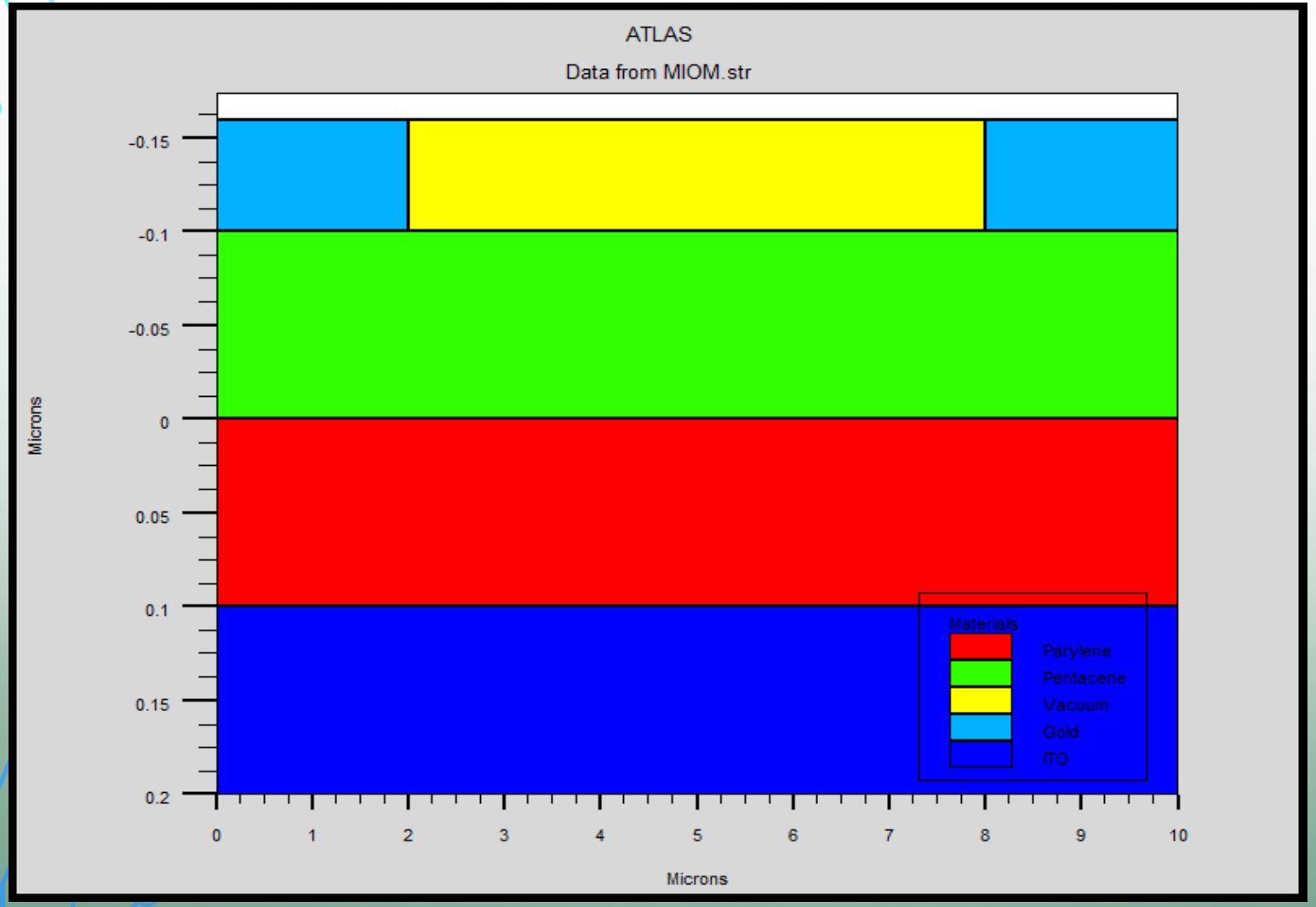
- Essentially a Field Effect Transistor with an organic semiconductor forming the channel and a dielectric forming the gate insulator
- Relatively less expensive to fabricate



ORGANIC FIELD EFFECT TRANSISTOR (OFET)

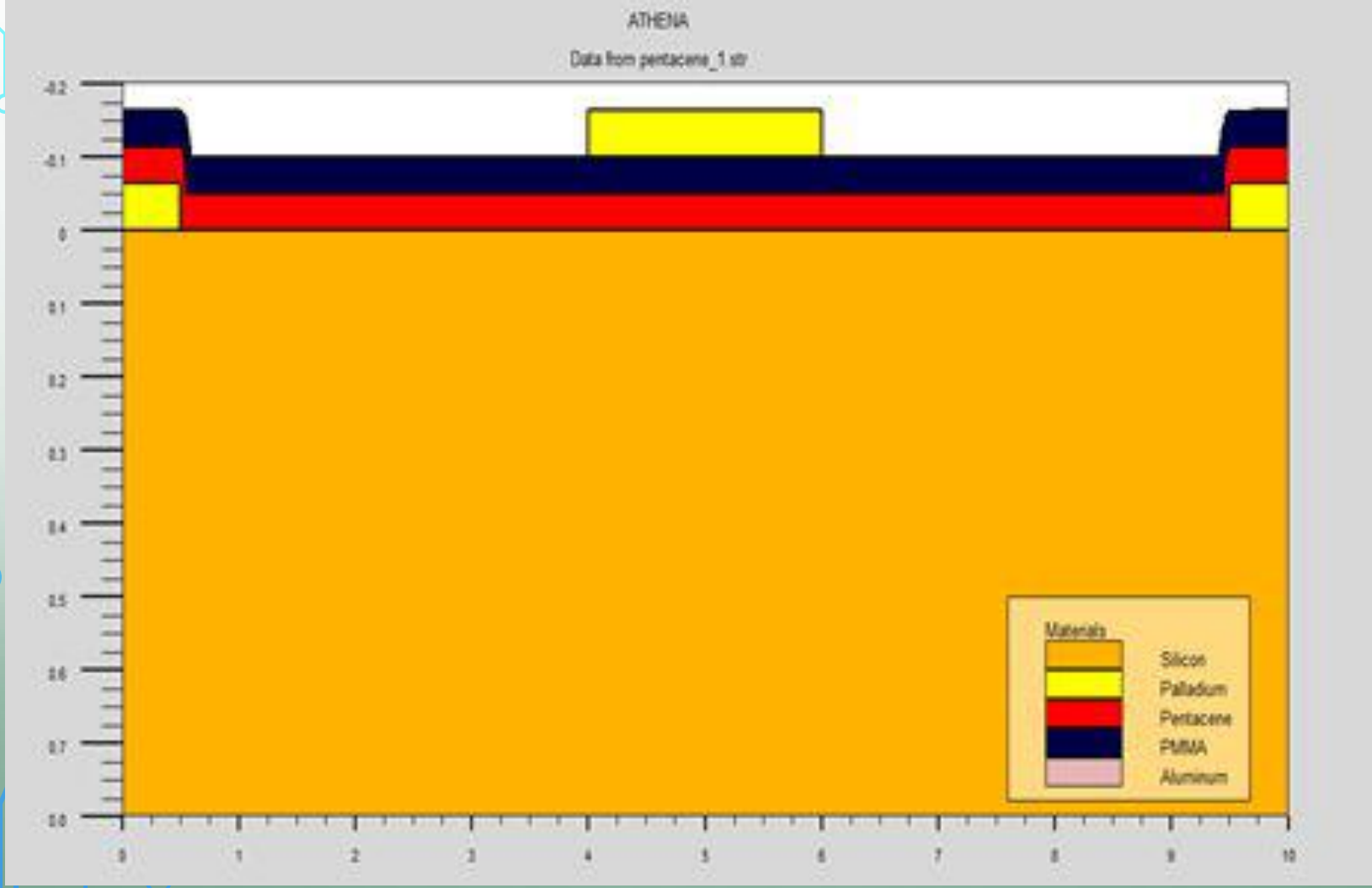
- First OFET: based on a polymer of thiophene molecules, a type of conjugated polymer (1987)
- One common feature of OFET materials is the inclusion of an aromatic or otherwise conjugated π-electron system, facilitating the delocalization of orbital wavefunctions
- Some active semiconducting layers: *rubrene, tetracene, pentacene, diindenoperylene, perylenediimides,* and polymers such as *polythiophenes, polyfluorene, polydiacetylene, poly(2,5-thienylene vinylene), poly(p-phenylene vinylene) (PPV).*

SIMULATIONS | DEVICE STRUCTURE



Layer	Thickness
ITO over Glass	100 nm
Parylene (Gate insulator)	100 nm
Pentacene (OSC)	100 nm
Gold	60 nm

SIMULATIONS | DEVICE STRUCTURE



Layer	Thickness
Pentacene (OSC)	450nm
PMMA (Gate insulator)	550nm
Palladium (Drain and Source contacts)	650nm

DEVICE SIMULATION TECHNIQUES

- ATLAS (Silvaco) in order to simulate the transistors (previously shown)
  - Predicts the electrical properties by solving the Poisson's and Continuity equations, and classical equations of carrier currents including drift and diffusion terms.

Poisson's Equation: $\mathrm{div}(\varepsilon \nabla \psi) = -\rho$

Carrier Continuity Equations: $\partial n/\partial t = (1/q)\mathrm{div}J_n + G_n - R_n$

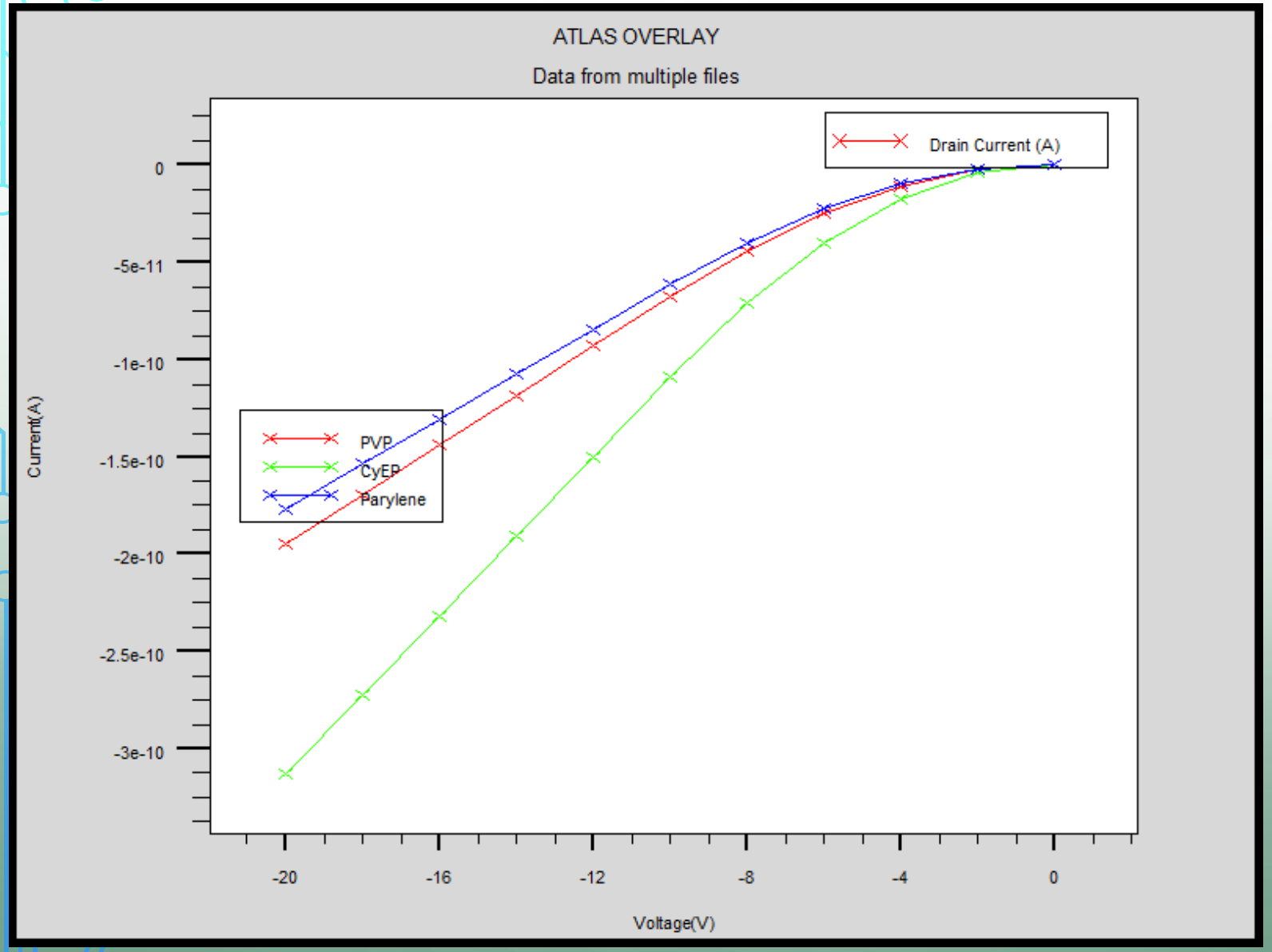
$\partial p/\partial t = (-1/q)\mathrm{div}J_p + G_p - R_p$

- Poole-Frenkel mobility model was used for the Pentacene conduction channel
- Support for organic devices is limited on ATLAS

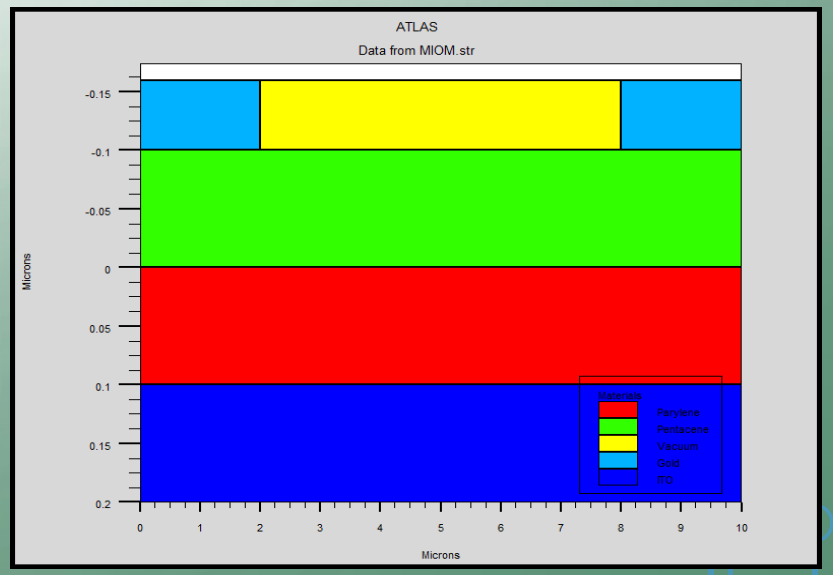
Silvaco TCAD

- One of the famous software used to simulate both processes and devices
- **ATHENA** is used as *process simulator* where as **ATLAS** is used as *device simulator*
- *ATHENA* - Etching and deposition, Implantation, Diffusion and Oxidation simulation capabilities
- *ATLAS* - Uses a tetrahedral meshing engine for fast and accurate device simulation of complex 3D geometries

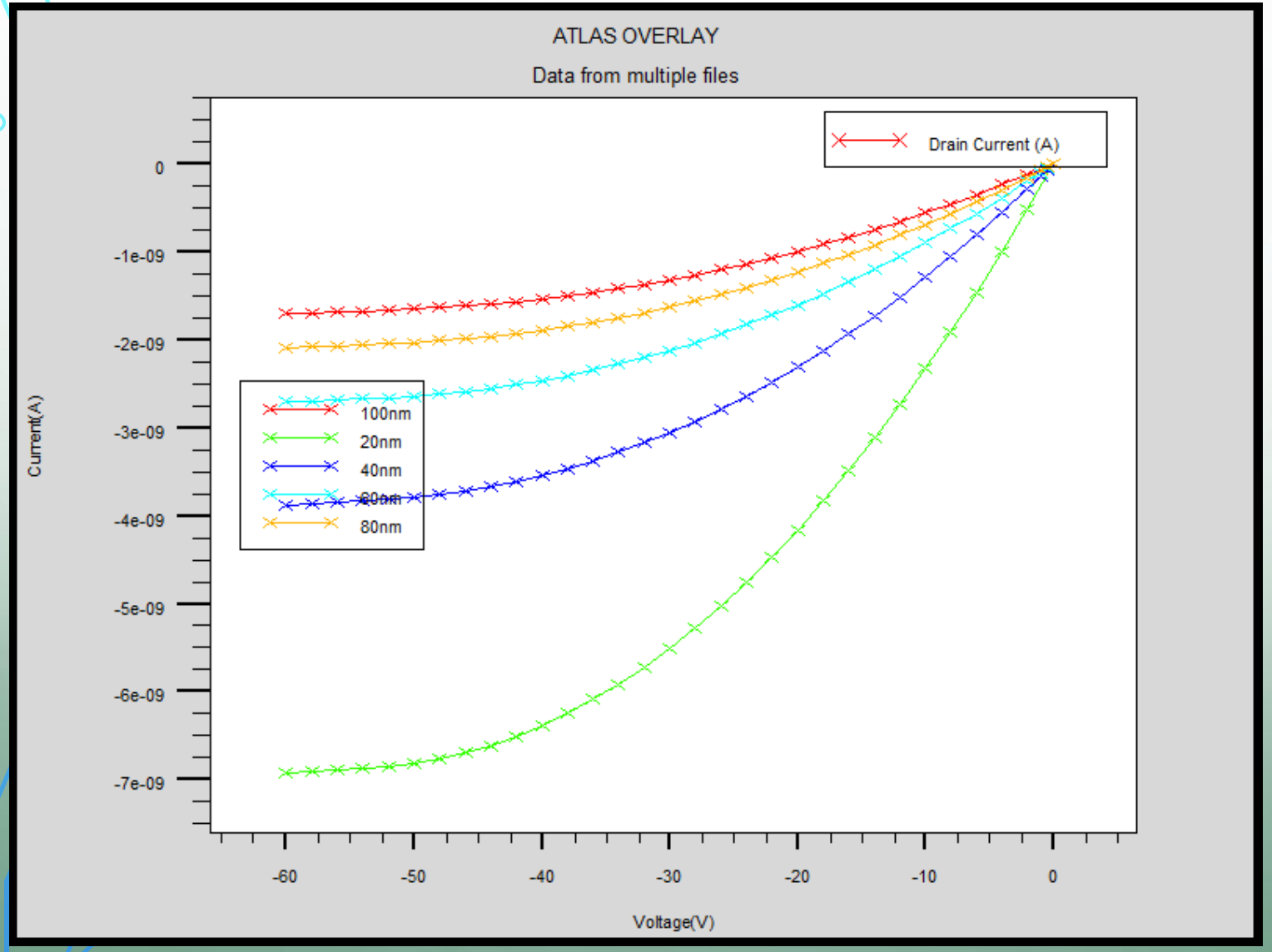
SIMULATION RESULTS – OFET-1



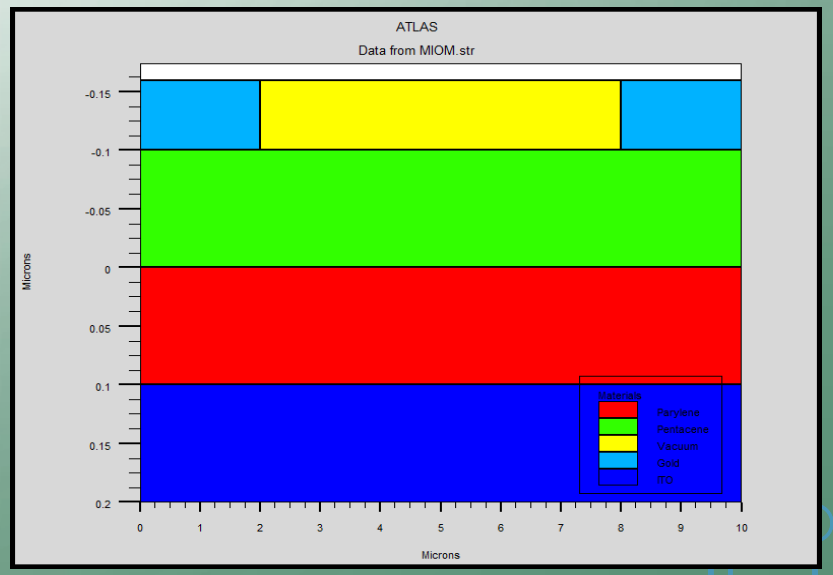
Transfer characteristics for the model in OFET-1 for Drain Voltage of -10V, for different Gate insulators



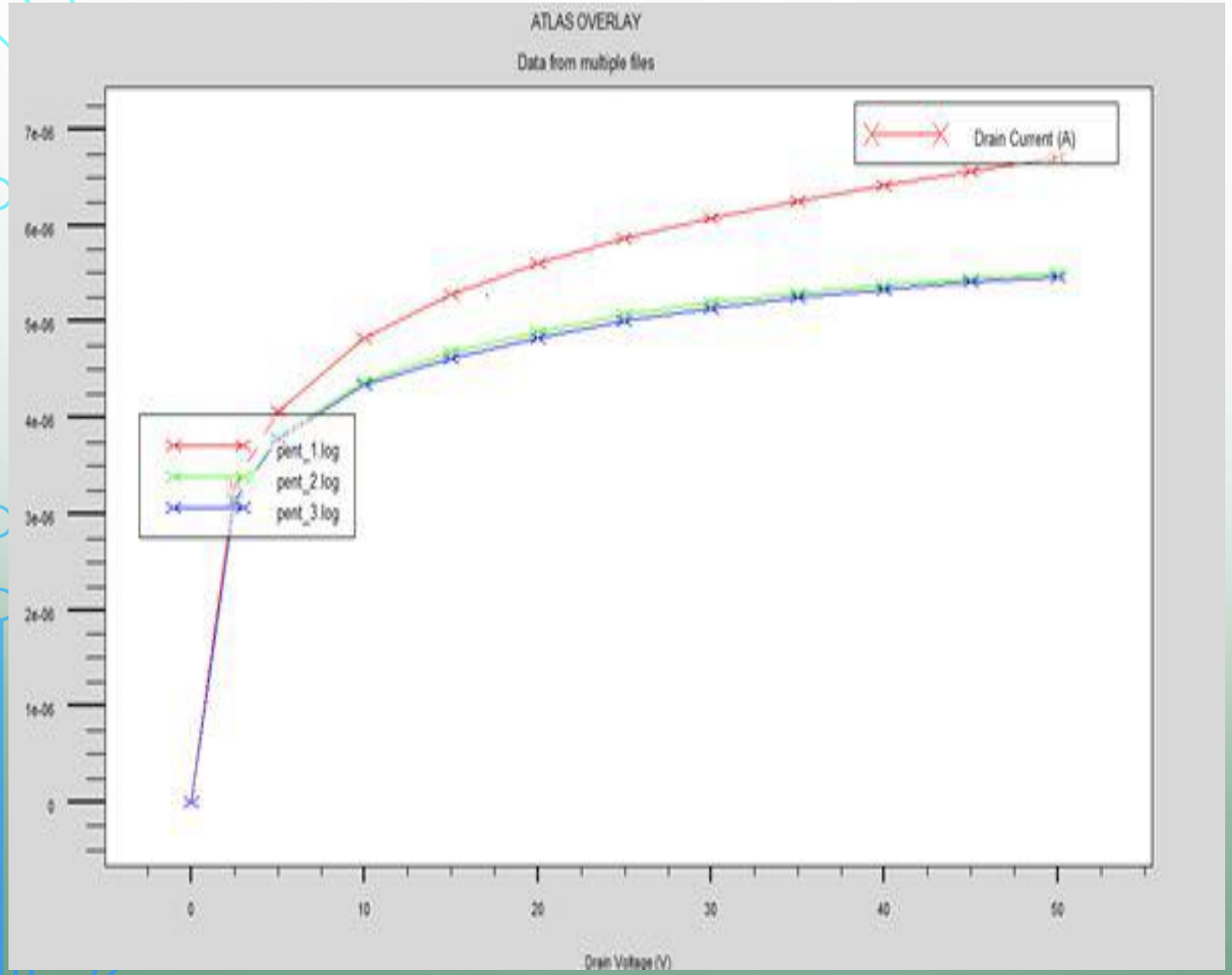
SIMULATION RESULTS – OFET-1



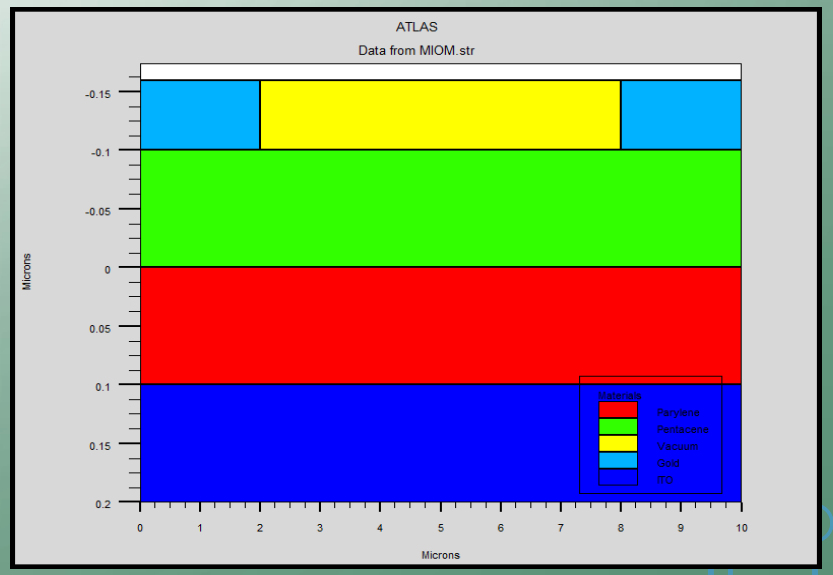
Output characteristics for the model in OFET-1 for Gate Voltage of -50V, for different Gate insulators



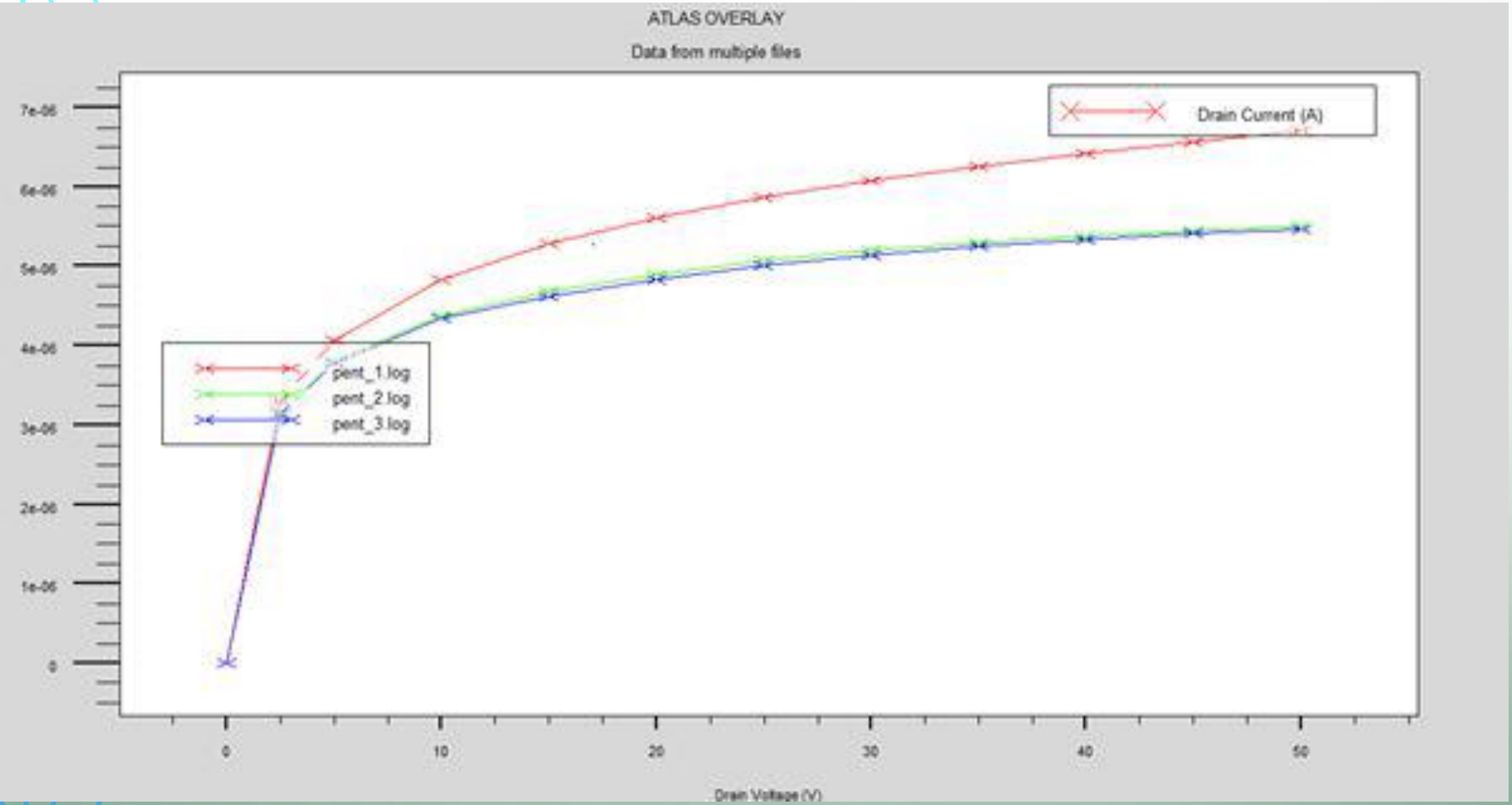
SIMULATION RESULTS – OFET-1



Output characteristics for the model in OFET-1 for different Gate Voltages, with PVP as gate insulator

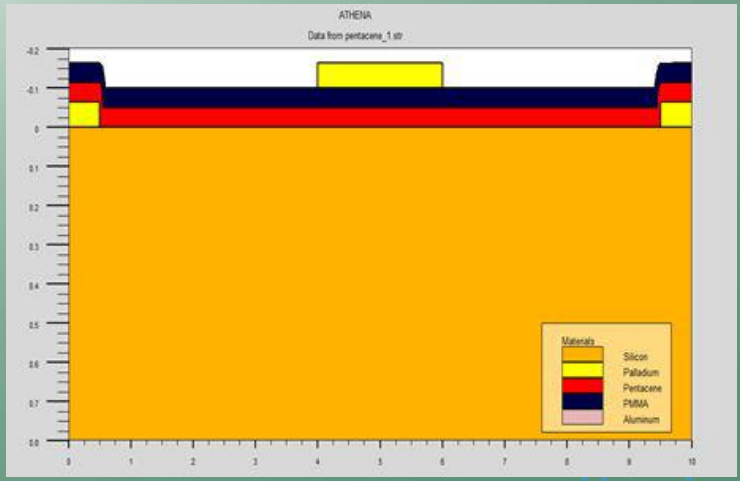


SIMULATION RESULTS – OFET-2



Drain characteristics for the model in OFET-2

- In this model, the saturation current came out to be $6x10^{-6}$A: close to the experimentally obtained value
- The PF mobility value was about $0.04cm^{-2}/Vs$, which was close to the reported value $0.069cm^{-2}/Vs$



SUMMARY

- As can be observed from the saturation current, field effect mobility of the above mentioned configurations, the Field Effect Transistor (FET) can be designed accordingly depending on the requirements by using various conducting polymers and gate insulators

- By using the suitable materials for gate insulators, active semiconductors and by adjusting the parameters such as thickness, we can get the required properties of the OFETs

- For the models we have worked on:
  The simulations in the OFET-2 model agree closely with the experimental results, but the simulation results for the model OFET-1 do not quite match with the results we expected

LIMITATIONS AND FUTURE WORK

- Poor electron mobility in n-type OSCs (Organic Semiconductors), limiting their effectiveness. One solution could be the cyanation of materials currently used in p-type OFETs thereby reducing the instability radical anions in air, in n-type OSCs
- Solution processible OSCs, the current methods of physical semiconductor deposition, such as vacuum sublimation and thermal evaporation, are less cost-effective than methods using soluble organic semiconductors. But due to less carrier mobility & on-off ratios, less desirable.
  Future work may attempt to find soluble OSCs with better electronic properties
- Problems related to the anisotropy inherent in most of the OSCs due to their linear nature
- Another area of related research is into the design of fully-organic FETs, in which all parts of the transistor are composed of organic materials, not just the semiconductor.
  Useful for cheap and flexible cicuits such asRFID tags, cheap electronic price tags and electronic paper.

ACKNOWLEDGEMENTS

- Our mentor Dr. E Varadarajan
- Dr. Sudipta Bhattacharya
- Research Innovation Center (RIC), DRDO
- Our faculty advisor: Dr. Somnath Chanda Roy
- Department of Physics, IIT Madras
- Department of Electrical Engineering, IIT Madras

THANK YOU